

Buy Side: Assignment 3

Financial Engineering

Covariance Estimation Techniques & Statistical Factor Models

Instructions

- **Delivery:** Friday 18:00 p.m. the 3rd of April to financial.engineering.polimi@gmail.com, with subject “BS: Assignment 3, Group #”;
- Deliver
 - a **short** pdf document with the results. List the errors you’ve found.
 - the code written in Python using your preferred IDE;
- Fill the gaps where appropriate and correct the errors (in case of incoherences between this document and the code, follow this document);
- Comment (in detail) the code and use explicative variable names;
- Use the market data provided in the files `sx5e_underlyings.csv` and `ticker_details.csv`.

Case study

A junior analyst working for the Equity Statistical Arbitrage pod of Polimi Capital Management is tasked with studying some novel techniques with two main preliminary objectives:

- Reduce the noise in the covariance matrix estimate
- Understand the pervasive sources of risk affecting stock returns

To conduct his research he focuses on the Euro Stoxx 50 constituents, performs monthly rebalancing (end of month) and at every rebalance uses the last two years of data. Assume total return indices in EUR and use simple returns. Ignore the risk-free rate.

Questions

1. Part I: Covariance Estimation
 - a. Consider the first full estimation window (i.e. the first rebalance date for which you have enough data) and calibrate the risk aversion parameter for the mean-variance portfolio to target a gross exposure of approximately 3. For the remainder of the analysis use the calibrated risk aversion parameter;
 - b. Implement the shrinkage target and relative intensity based on a constant correlation model;
 - c. Implement the shrinkage target and relative intensity based on a single factor model;
 - d. Compute the minimum variance and the mean-variance portfolios for the sample covariance matrix and for the two shrunk matrices;
 - e. Discuss the similarities and differences between the two shrinkage estimators: what are their different rationales? Focus on the evolution of the shrinkage intensities, on the comparison of the condition number of the sample and the shrunk covariance matrices and on the out of sample characteristics of the two portfolios (volatility, performance, turnover and gross exposure);
2. Part II: Statistical Factor Model
 - a. Implement a principal component analysis of the sample covariance matrix;

- b. Discuss the results, focusing on the factors turnover (as expressed by the cosine similarity between successive principal components) and the percentage of variance explained by the various factors in terms of the market condition. How is factor turnover related to the time window used to compute the covariance matrix? Does using the correlation matrix instead of the covariance matrix affect the results?
 - c. Can you give an interpretation of the first few (say three) factors on January 31st, 2019? (Hint: use sectors data);
3. Part III: Connecting the Dots (optional)
- a. Keeping as reference January 31st 2019, discuss the connection between shrinkage methods and PCA-based denoising of the covariance matrix studying the behavior of the eigenvalue spectrum of the two methods. Let the number of retained principal components vary in $\{1, 3, 5, 10\}$;
 - b. The dataset used in the exercise does not fully reveal the merits of shrinkage due to the limited cross-sectional dimension. Explain why this is the case.